

Module 7: Classification

Part 3: naive Bayes

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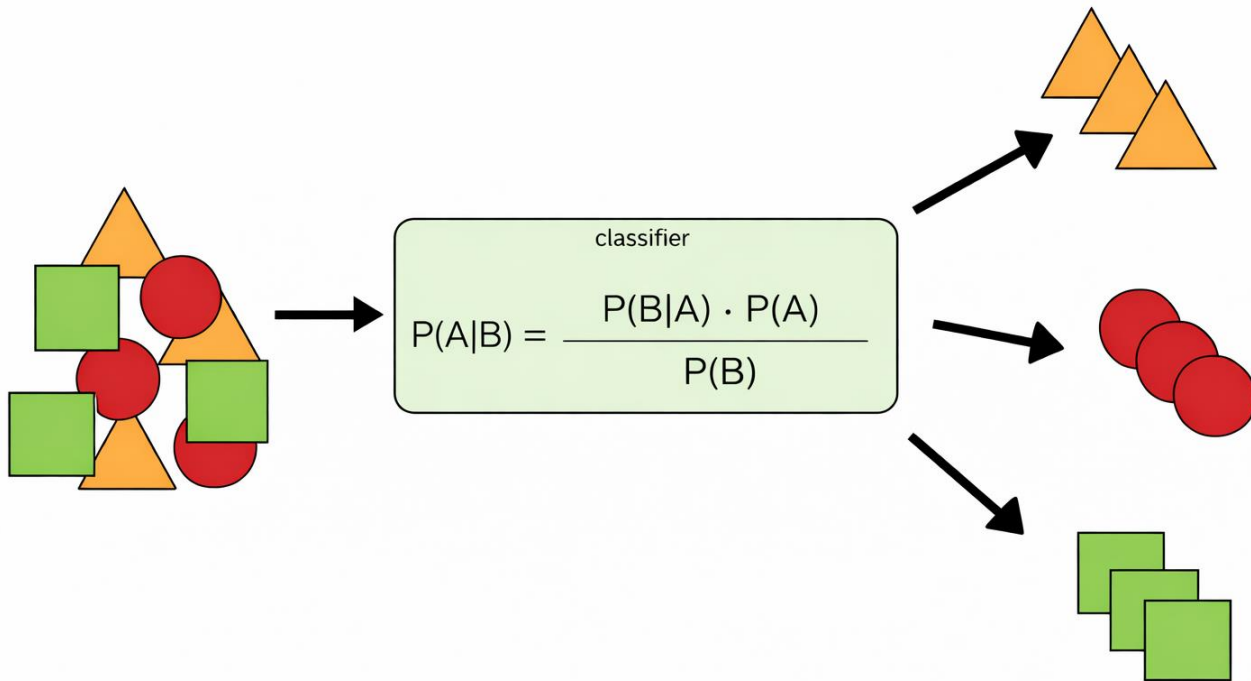
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Some lecture slides and instructional materials in this course are adapted from the following sources:

<https://statquest.org/index.html>

Naive Bayes Classifier



Naive Bayes

Recall that in LDA/QDA, we used Bayes Theorem to approximate $P(Y = k | X)$

$$P(Y = k | X) = \frac{P(X | Y = k)P(Y = k)}{P(X)}$$

- $P(X | Y = k) = f_k(\mathbf{x})$ is the p -dimensional density function for an observation in the k th class.
- Estimating this density function is VERY challenging.
- LDA/QDA makes strong assumptions about this density function. This makes a very challenging problem much simpler.
- Naive Bayes makes a different attempt to estimate $P(X | Y = k)$.

Naive Bayes' Assumption

- Within the k th class, the p predictors are independent.
- Recall, we denote $P(X | Y = k) = f_k(x)$ where $x = (x_1, x_2, \dots, x_p)$.

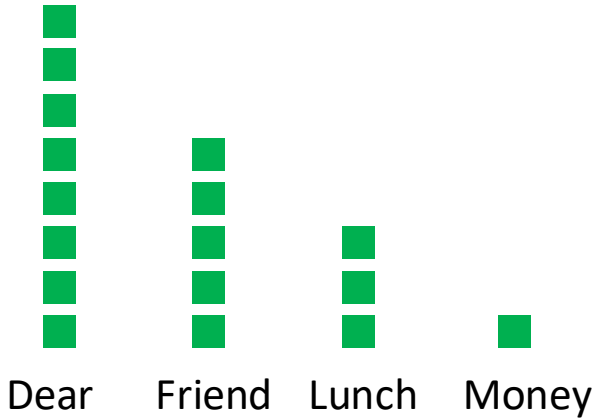
Mathematically, this means for $k = 1, \dots, K$:

$$f_k(x_1, x_2, x_3, \dots, x_p) = f_{k1}(x_1) \times f_{k1}(x_2) \times \dots \times f_{kp}(x_p)$$

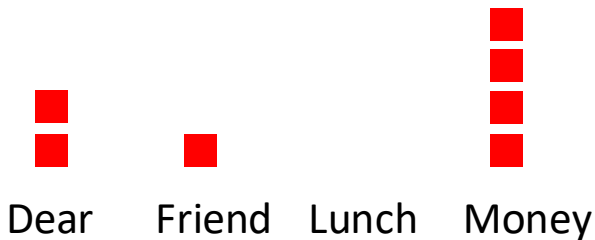
Spam Email Classifier

Prior Probability (Initial guess)

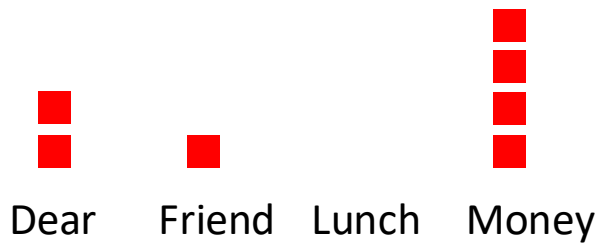
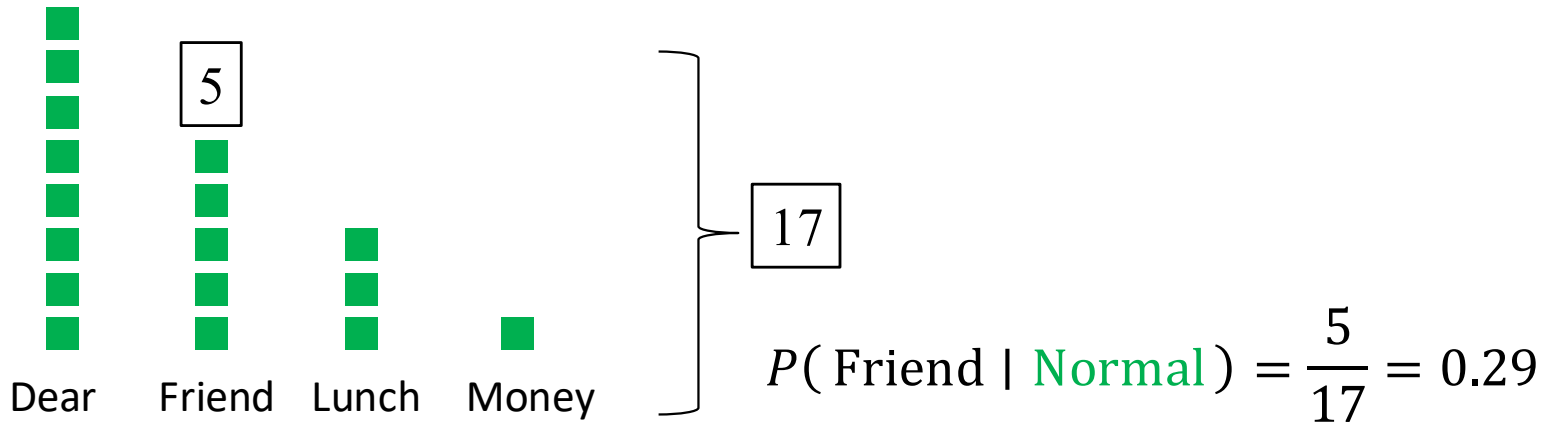
$$P(\text{Normal}) = \frac{8}{8 + 4} = 0.67$$



$$P(\text{Spam}) = \frac{4}{8 + 4} = 0.33$$

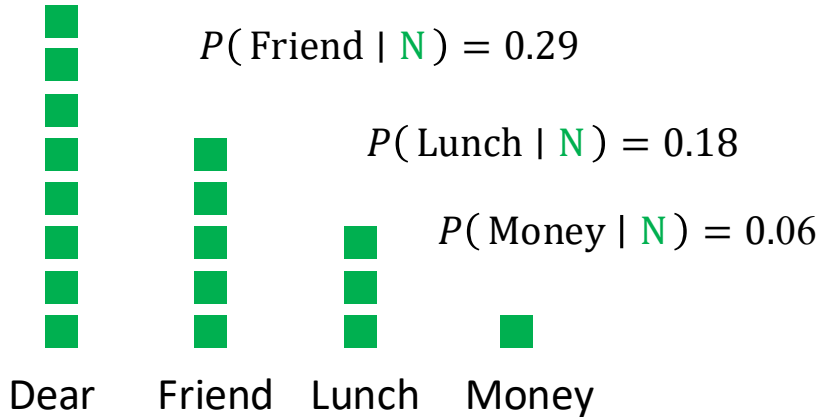


Spam Email Classifier

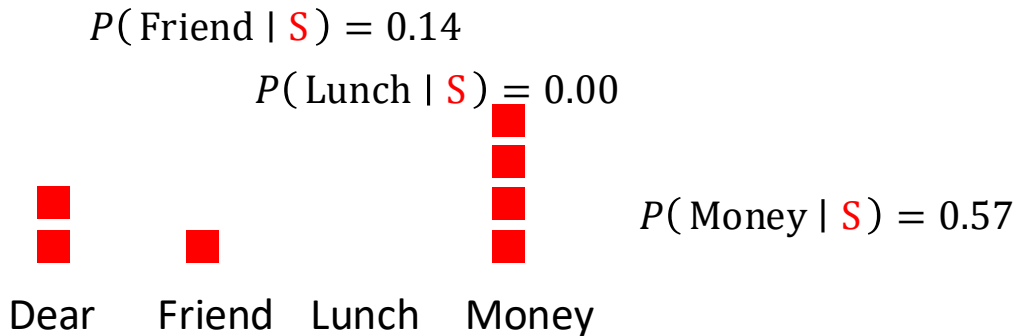


Spam Email Classifier

$$P(\text{Dear} \mid \text{N}) = 0.47$$



$$P(\text{Dear} \mid \text{S}) = 0.29$$



Spam Email Classifier

$$\begin{aligned}P(\text{Dear} \mid \text{N}) &= 0.47 \\P(\text{Friend} \mid \text{N}) &= 0.29 \\P(\text{Lunch} \mid \text{N}) &= 0.18 \\P(\text{Money} \mid \text{N}) &= 0.06\end{aligned}$$

Email: Dear Friend
↳ normal email ($0.091 > 0.013$)

$$\begin{aligned}&P(\text{N}) \times P(\text{Dear} \mid \text{N}) \times P(\text{Friend} \mid \text{N}) \\&= 0.67 \cdot 0.47 \cdot 0.29 \\&= 0.091\end{aligned}$$

$$\begin{aligned}P(\text{Dear} \mid \text{S}) &= 0.29 \\P(\text{Friend} \mid \text{S}) &= 0.14 \\P(\text{Lunch} \mid \text{S}) &= 0.00 \\P(\text{Money} \mid \text{S}) &= 0.57\end{aligned}$$

$$\begin{aligned}&P(\text{S}) \times P(\text{Dear} \mid \text{S}) \times P(\text{Friend} \mid \text{S}) \\&= 0.33 \cdot 0.29 \cdot 0.14 \\&= 0.013\end{aligned}$$

Spam Email Classifier

$$P(\text{Normal}) = 0.67$$

$$P(\text{Dear} | \text{N}) = 0.47$$

$$P(\text{Friend} | \text{N}) = 0.29$$

$$P(\text{Lunch} | \text{N}) = 0.18$$

$$P(\text{Money} | \text{N}) = 0.06$$

Email: Lunch Money Money Money Money

$$P(\text{N}) \times P(\text{Money} | \text{N}) \times P(\text{Money} | \text{N}) \\ \times P(\text{Money} | \text{N}) \times P(\text{Money} | \text{N})$$

$$= 0.67 \cdot [0.06]^4 \cdot 0.18$$

=

$$P(\text{Spam}) = 0.33$$

$$P(\text{Dear} | \text{S}) = 0.29$$

$$P(\text{Friend} | \text{S}) = 0.14$$

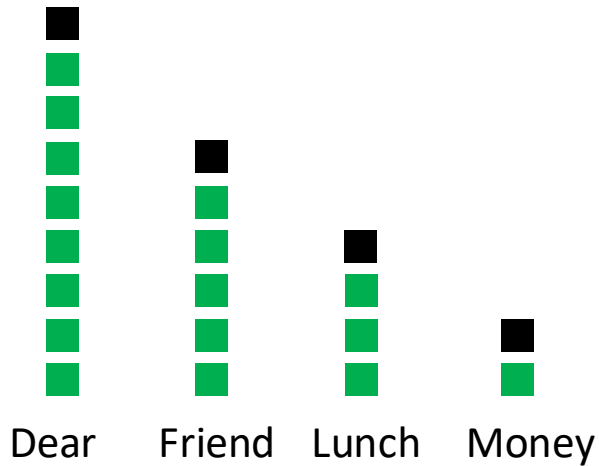
$$P(\text{Lunch} | \text{S}) = 0.00$$

$$P(\text{Money} | \text{S}) = 0.57$$

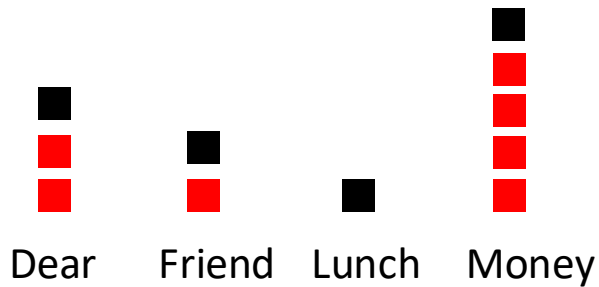
$$P(\text{S}) \times P(\text{Money} | \text{S}) \times P(\text{Money} | \text{S}) \\ \times P(\text{Money} | \text{S}) \times P(\text{Money} | \text{S})$$

$$= 0.33 \cdot [0.57]^4 \cdot 0 = 0$$

Spam Email Classifier



Black box: Add 1 count



Spam Email Classifier

$$P(\text{Normal}) = 0.67$$

$$P(\text{Dear} | \text{N}) = 0.43$$

$$P(\text{Friend} | \text{N}) = 0.29$$

$$P(\text{Lunch} | \text{N}) = 0.19$$

$$P(\text{Money} | \text{N}) = 0.10$$

Email: Lunch Money Money Money Money
→ spam email

$$P(\text{N}) \times P(\text{Money} | \text{N}) \times P(\text{Money} | \text{N}) \\ \times P(\text{Money} | \text{N}) \times P(\text{Money} | \text{N})$$

$$= 0.43 \cdot [0.1]^4 \cdot 0.19$$

$$= 8.17e^{-6}$$

$$P(\text{Spam}) = 0.33$$

$$P(\text{Dear} | \text{S}) = 0.27$$

$$P(\text{Friend} | \text{S}) = 0.18$$

$$P(\text{Lunch} | \text{S}) = 0.09$$

$$P(\text{Money} | \text{S}) = 0.45$$

$$P(\text{S}) \times P(\text{Money} | \text{S}) \times P(\text{Money} | \text{S}) \\ \times P(\text{Money} | \text{S}) \times P(\text{Money} | \text{S})$$

$$= 0.33 \cdot [0.45]^4 \cdot 0.09$$

$$= 0.0012$$

Spam Email Classifier

Assumption

- All words are independent
- Ignore relationships between words

Naive Bayes' Assumption

- Do we really believe this assumption? No.
- Even though this is a highly naive assumption, it often leads to relatively decent results.
- From the viewpoint of the bias-variance tradeoff: Naive Bayes introduces some bias, but reduces variance, leading to a classifier that actually works quite well in practice.

What does the problem boil down to?

$$\begin{aligned} P(Y = k | X) &= \frac{P(X | Y = k)P(Y = k)}{P(X)} \\ &= \frac{f_k(X)\pi_k}{P(X)} \\ &= \frac{f_{k1}(x_1) \times f_{k1}(x_2) \times \cdots \times f_{kp}(x_p)\pi_k}{P(X)} \end{aligned}$$

Options for estimating these marginal densities f_{kj} :

- If the predictor is **qualitative**, count the proportion of trainings observations for the predictor corresponding to each class.
- If the predictor is **quantitative**, use a parametric approach (if we know its distribution) or a non-parametric approach (kernel density estimation).

Naive Bayes

- Trades one strong assumption for another.
- If you have qualitative predictors, Naive Bayes is a good choice (because normality assumption definitely does not hold).
- When n is not large enough relative to p , Naive Bayes can also outperform LDA/QDA.

Implementation

- See R script: naivebayes.R